192325023-Y.Harika

R Programming Lab – Assignment 2 (CO2)

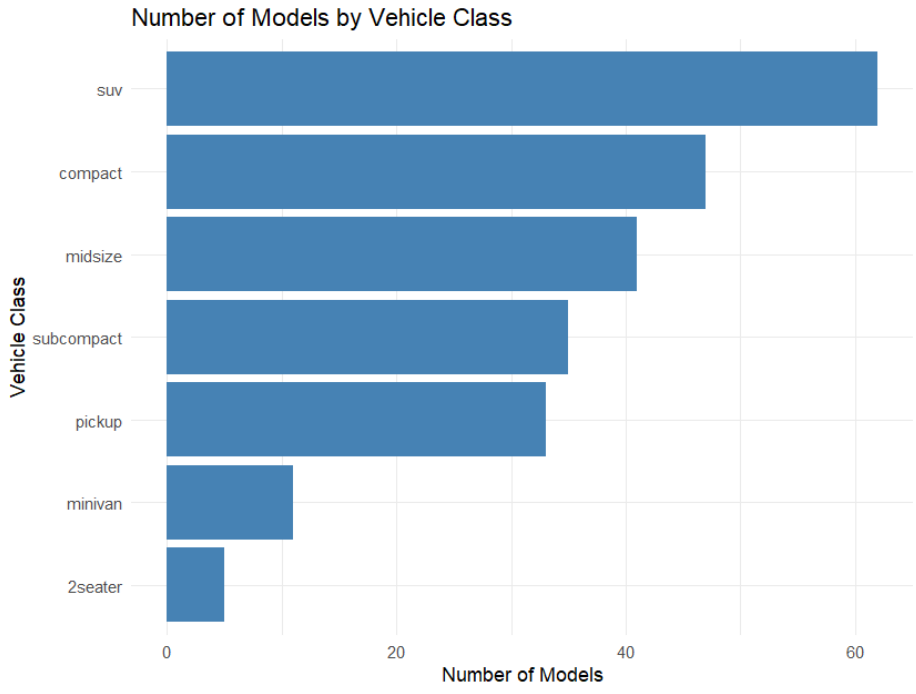
Title: Applying Visualization Methods for Amounts, Distributions, Proportions and Relationships

Packages required: ggplot2, dplyr, forcats, patchwork, treemapify

Q1. Visualizing Amounts

(a) Horizontal bar chart of models in each vehicle class [4 marks]

```
library(ggplot2)
library(dplyr)
class_counts <- mpg %>%
  count(class, sort = TRUE)
ggplot(class_counts, aes(x = reorder(class, n), y = n)) +
  geom_col(fill = "steelblue") +
  coord_flip() +
  labs(title = "Number of Models by Vehicle Class",
       x = "Vehicle Class", y = "Number of Models") +
  theme_minimal()
```



Interpretation: The horizontal bar chart compares the number of vehicle models in each class. Classes are sorted from most to least frequent, making the most common class easy to identify.

b) Grouped bar chart by class and drive type [3 marks]

```
ggplot(mpg, aes(x = class, fill = drv)) +
  geom_bar(position = "dodge") +
  labs(title = "Vehicle Counts by Class and Drive Type",
       x = "Vehicle Class", y = "Count",
       fill = "Drive Type") +
  theme_minimal()
```

Interpretation: Each vehicle class is divided into side-by-side bars for drive types (4, f, and r). This allows comparison of drive-type counts within and across classes.

(c) Stacked bar chart of diamond counts by cut and clarity [3 marks]

```
library(ggplot2)

ggplot(diamonds, aes(x = cut, fill = clarity)) +
  geom_bar() +
  labs(title = "Diamond Counts by Cut and Clarity",
       x = "Cut", y = "Count", fill = "Clarity") +
  theme_minimal()
```

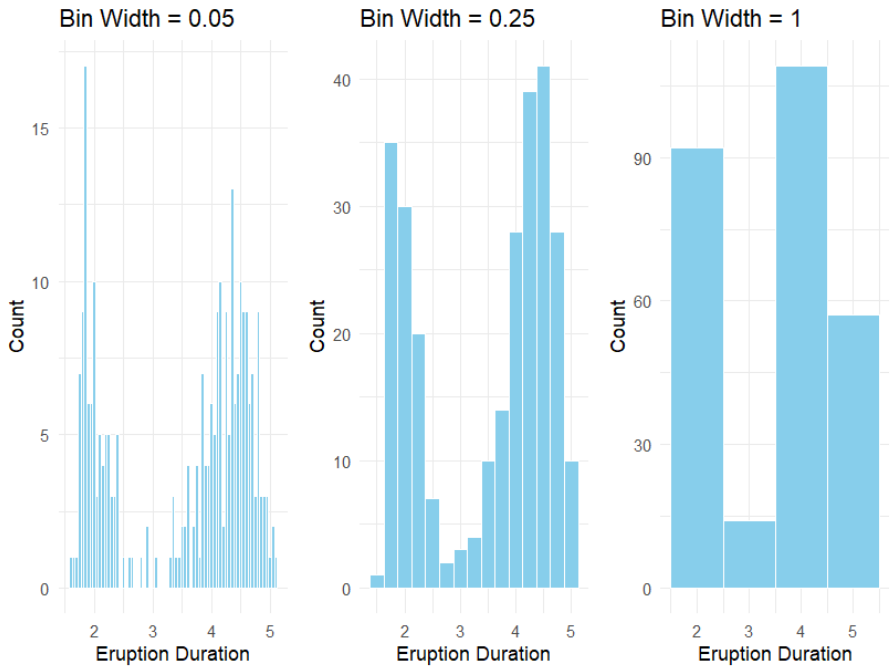


Interpretation: The total height represents the number of diamonds in each cut category. Colored segments show how clarity grades contribute to each cut total.

Q2. Visualizing a Single Distribution

(a) Histograms of faithful eruption duration with three bin widths [5 marks]

```
library(ggplot2)
library(patchwork)
p1 <- ggplot(faithful, aes(x = eruptions)) +
  geom_histogram(binwidth = 0.05, fill = "skyblue", color = "white") +
  labs(title = "Bin Width = 0.05", x = "Eruption Duration", y = "Count") +
  theme_minimal()
p2 <- ggplot(faithful, aes(x = eruptions)) +
  geom_histogram(binwidth = 0.25, fill = "skyblue", color = "white") +
  labs(title = "Bin Width = 0.25", x = "Eruption Duration", y = "Count") +
  theme_minimal()
p3 <- ggplot(faithful, aes(x = eruptions)) +
  geom_histogram(binwidth = 1, fill = "skyblue", color = "white") +
  labs(title = "Bin Width = 1", x = "Eruption Duration", y = "Count") +
  theme_minimal()
p1 + p2 + p3
```

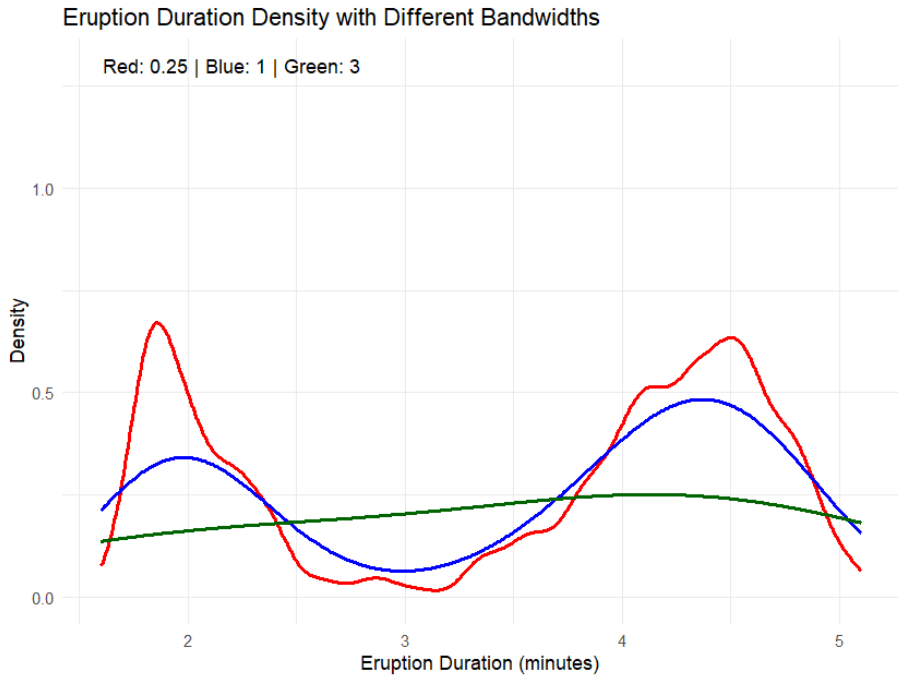


Interpretation: A very small bin width (0.05) produces a detailed but potentially noisy histogram. A width of 0.25 shows the overall shape more clearly. A width of 1 is coarse and may hide peaks or gaps. Bin width controls the trade-off between detail and smoothness.

(b) Kernel density estimates with bandwidth adjustments

```
ggplot(faithful, aes(x = eruptions)) +
  geom_density(adjust = 0.25, color = "red", linewidth = 1) +
  geom_density(adjust = 1, color = "blue", linewidth = 1) +
  geom_density(adjust = 3, color = "darkgreen", linewidth = 1) +
```

```
labs(title = "Eruption Duration Density with Different Bandwidths",
     x = "Eruption Duration (minutes)", y = "Density",
     color = "Bandwidth") +
theme_minimal() +
annotate("text", x = 2.2, y = 1.3, label = "Red: 0.25 | Blue: 1 | Green: 3")
```

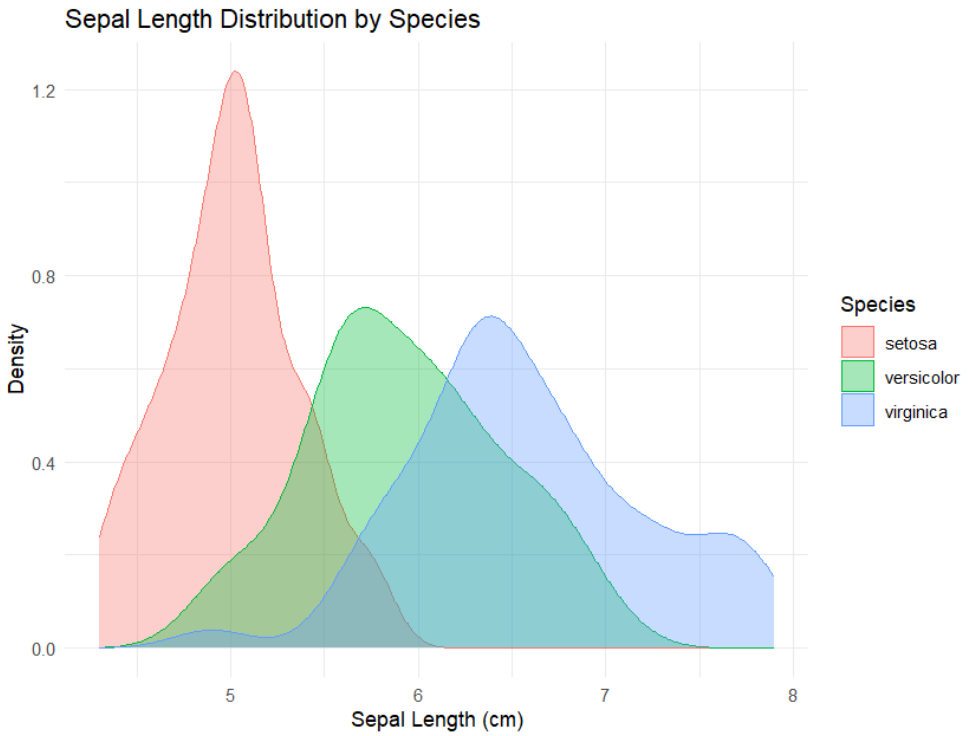


Interpretation: A small adjustment (0.25) creates a highly detailed, rough density curve. The default adjustment (1) balances detail and smoothness. A large adjustment (3) smooths the curve substantially and may merge distinct modes.

Q3. Visualizing Multiple Distributions

(a) Overlapping density plots of Sepal.Length by species

```
ggplot(iris, aes(x = Sepal.Length, fill = Species,
                 color = Species)) +
  geom_density(alpha = 0.35) +
  labs(title = "Sepal Length Distribution by Species",
       x = "Sepal Length (cm)", y = "Density",
       fill = "Species", color = "Species") +
  theme_minimal()
```

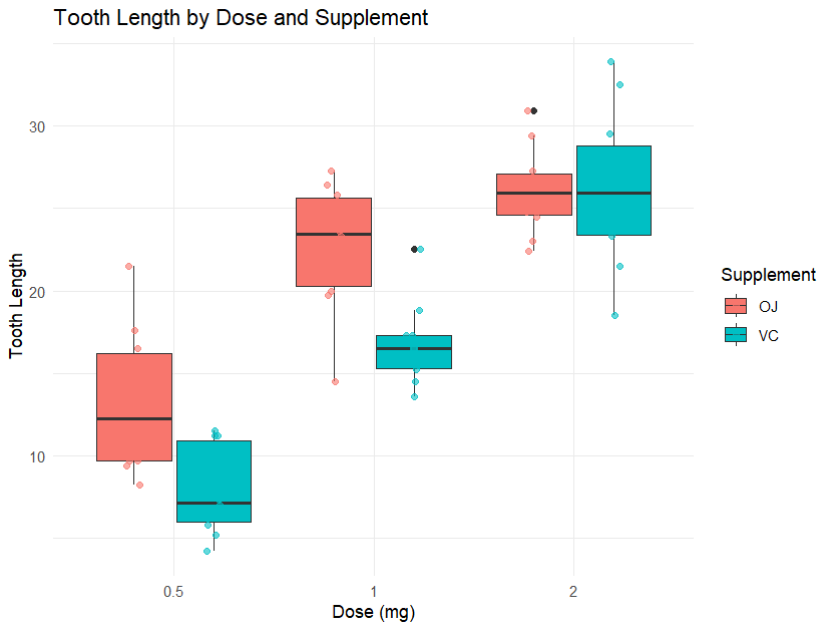


Interpretation: The overlapping density curves compare sepal-length distributions across the three species. Transparency makes overlapping regions visible and highlights differences in distribution shape and location.

(b) ToothGrowth boxplots by dose and supplement with jitter

```
data("ToothGrowth")

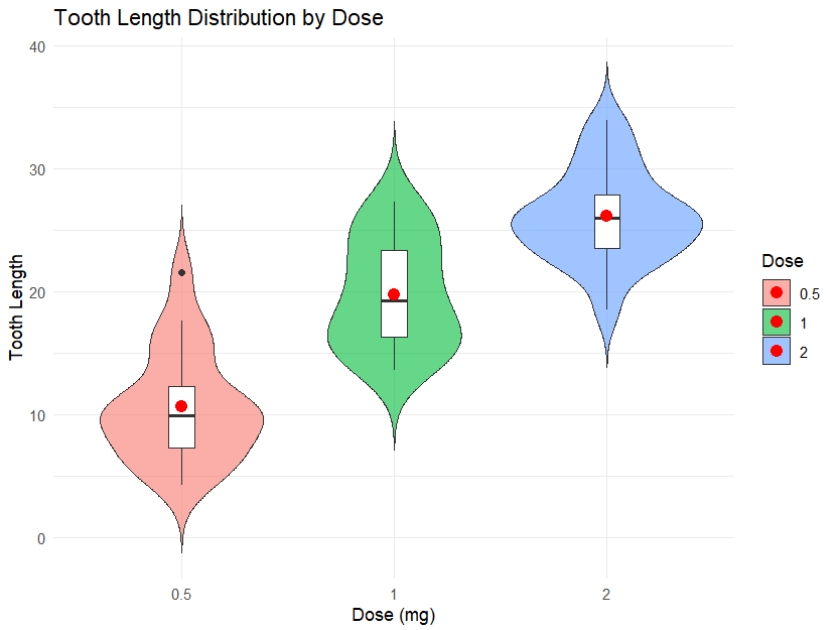
ggplot(ToothGrowth, aes(x = factor(dose), y = len,
                        fill = supp)) +
  geom_boxplot(position = position_dodge(width = 0.8)) +
  geom_jitter(aes(color = supp),
              position = position_jitterdodge(
                jitter.width = 0.15, dodge.width = 0.8),
              alpha = 0.6, size = 1.5) +
  labs(title = "Tooth Length by Dose and Supplement",
       x = "Dose (mg)", y = "Tooth Length",
       fill = "Supplement", color = "Supplement") +
  theme_minimal()
```



Interpretation: Boxplots summarize tooth-length distributions at each dose for each supplement type. Jittered points show individual observations and help reveal sample size, spread, and possible outliers.

(c) Violin plots with narrow boxplot and red mean [3 marks]

```
ggplot(ToothGrowth, aes(x = factor(dose), y = len,
                        fill = factor(dose))) +
  geom_violin(trim = FALSE, alpha = 0.6) +
  geom_boxplot(width = 0.12, fill = "white") +
  stat_summary(fun = mean, geom = "point",
               color = "red", size = 3) +
  labs(title = "Tooth Length Distribution by Dose",
       x = "Dose (mg)", y = "Tooth Length",
       fill = "Dose") +
  theme_minimal()
```



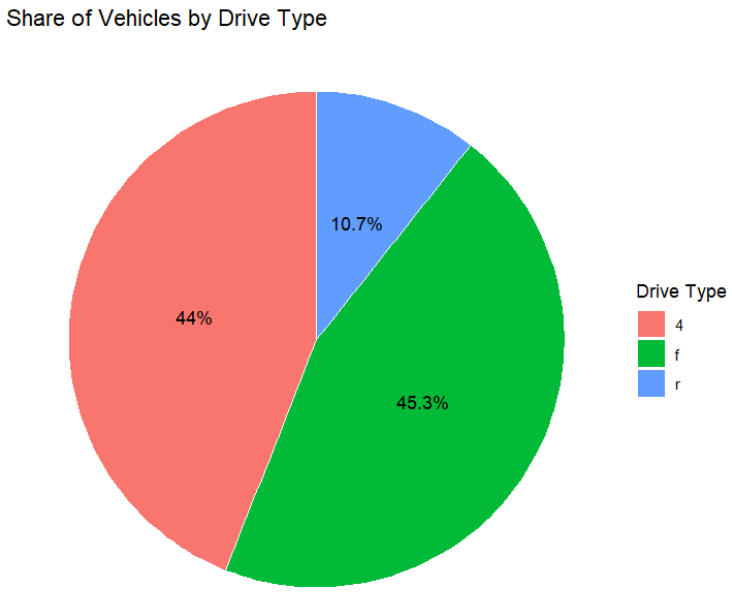
Interpretation: Violin width indicates the density of observations. The narrow boxplot summarizes the median and quartiles, while the red point marks the mean tooth length for each dose.

Q4. Visualizing Proportions [10 Marks]

(a) Pie chart of vehicle share by drive type with percentages [4 marks]

```
library(dplyr)
library(ggplot2)
drv_share <- mpg %>%
  count(drv) %>%
  mutate(percent = n / sum(n) * 100,
         label = paste0(drv, " (", round(percent, 1), "%)"))

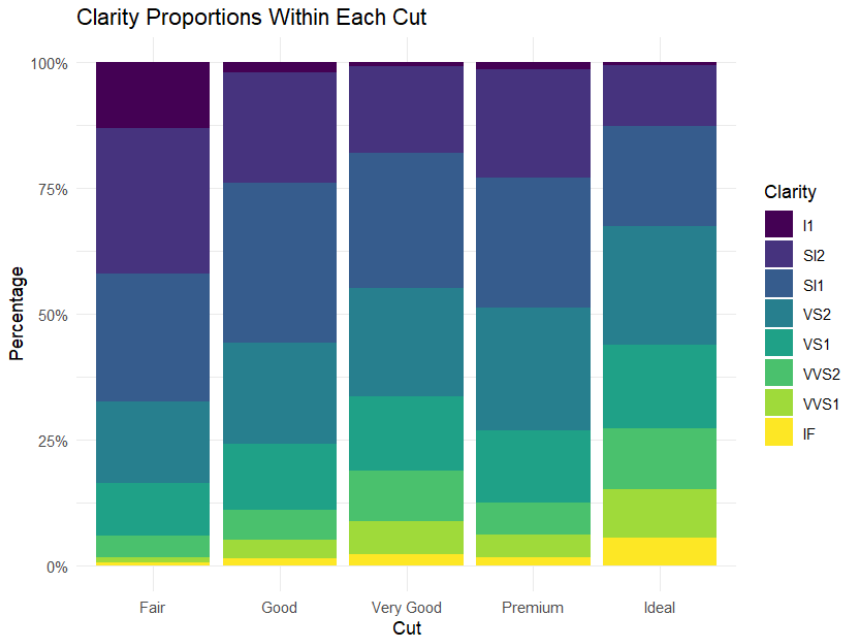
ggplot(drv_share, aes(x = "", y = n, fill = drv)) +
  geom_col(width = 1, color = "white") +
  coord_polar(theta = "y") +
  geom_text(aes(label = paste0(round(percent, 1), "%")),
            position = position_stack(vjust = 0.5)) +
  labs(title = "Share of Vehicles by Drive Type",
       fill = "Drive Type") +
  theme_void()
```



Interpretation: Each pie slice represents the percentage of vehicles belonging to a drive type. Percentage labels make the relative share of 4-wheel, front-wheel, and rear-wheel drive vehicles directly readable.

(b) 100% stacked bar chart of clarity within cut [3 marks]

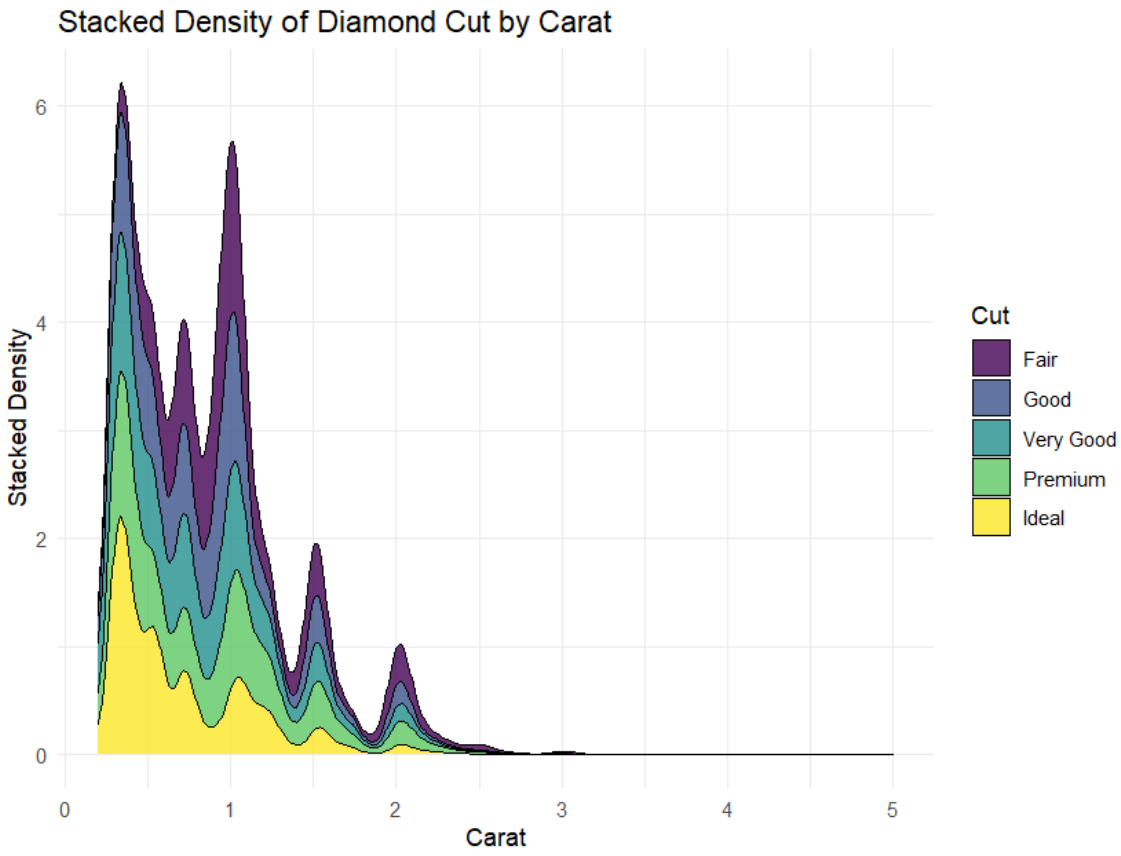
```
ggplot(diamonds, aes(x = cut, fill = clarity)) +
  geom_bar(position = "fill") +
  scale_y_continuous(labels = scales::percent) +
  labs(title = "Clarity Proportions Within Each Cut",
       x = "Cut", y = "Percentage", fill = "Clarity") +
  theme_minimal()
```



Interpretation: Every cut category has a bar of equal height (100%). The segment sizes show the proportion of each clarity grade within that cut, enabling comparison of composition rather than raw counts.

(c) Stacked density plot showing cut proportions along carat [3 marks]

```
ggplot(diamonds, aes(x = carat, fill = cut)) +
  geom_density(position = "stack", alpha = 0.8) +
  labs(title = "Stacked Density of Diamond Cut by Carat",
       x = "Carat", y = "Stacked Density",
       fill = "Cut") +
  theme_minimal()
```



Interpretation: The stacked density plot displays the distribution of carat values for each cut, with density contributions stacked. It helps compare how cut categories are distributed across carat sizes; the stacked heights should be interpreted as combined density, not direct percentages at each carat.

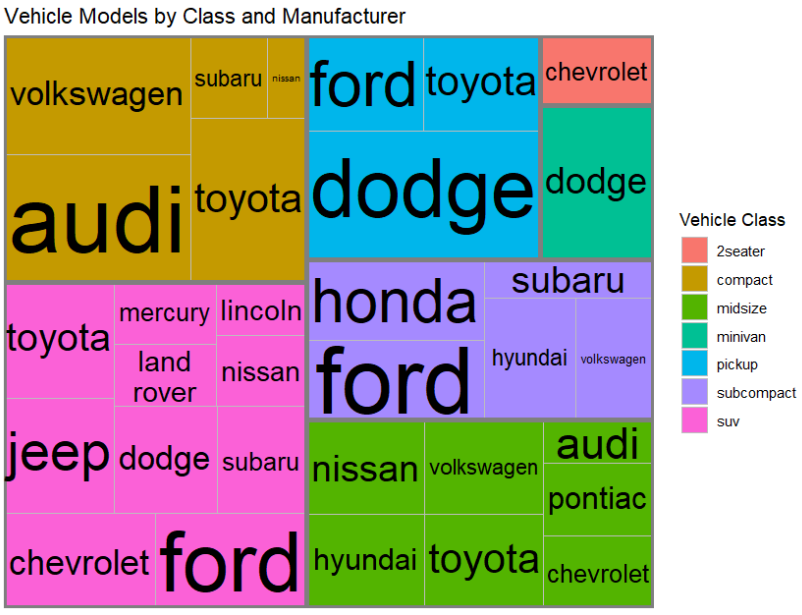
Q5. Nested Proportions and x–y Relationships [10 Marks]

(a) Treemap of vehicle class and manufacturer [5 marks]

```
library(ggplot2)
library(dplyr)
library(treemapify)

mpg_tree <- mpg %>%
  count(class, manufacturer, name = "models")

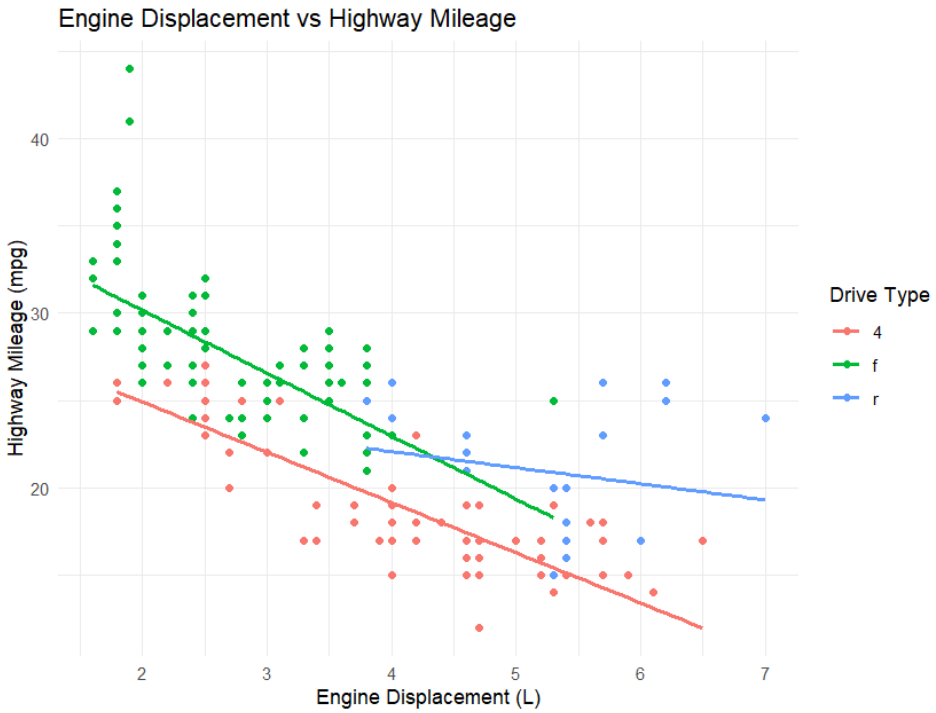
ggplot(mpg_tree,
       aes(area = models, fill = class,
           subgroup = class, label = manufacturer)) +
  geom_treemap() +
  geom_treemap_subgroup_border() +
  geom_treemap_text(
    aes(label = manufacturer),
    place = "centre", grow = TRUE, reflow = TRUE,
    min.size = 3
  ) +
  labs(title = "Vehicle Models by Class and Manufacturer",
       fill = "Vehicle Class") +
  theme_minimal()
```



Interpretation: Each rectangle represents a manufacturer nested within a vehicle class. Rectangle area corresponds to the number of models, while color identifies the class. Larger rectangles indicate manufacturers with more models in the dataset.

(b) Scatterplot of displacement vs highway mileage, colored by drive type, with regression lines [5 marks]

```
ggplot(mpg, aes(x = displ, y = hwy, color = drv)) +
  geom_point() +
  geom_smooth(method = "lm", se = FALSE) +
  labs(title = "Engine Displacement vs Highway Mileage",
       x = "Engine Displacement (L)",
       y = "Highway Mileage (mpg)",
       color = "Drive Type") +
  theme_minimal()
```



Interpretation: The scatterplot shows the relationship between engine displacement and highway mileage. Separate linear regression lines summarize the trend for each drive type. The overall pattern generally indicates that larger engines tend to have lower highway mileage, though the strength and slope may differ among drive types.